

K Ujjwal Reddy

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OBJECTIVE

An undergraduate background in Artificial Intelligence and Machine Learning, combined with hands-on projects such as Genetic Algorithm-Based Path Planning and HyperForge, provides a strong foundation in optimization, intelligent systems, and high-performance computing. Aspiring to pursue a Master's in Computer Science with a focus on Systems and Machine Intelligence. Particular interest lies in exploring autonomous systems, scalable algorithms, and adaptive computing frameworks, with the aim of contributing to research that bridges efficient system design and intelligent decision-making. The long-term goal is to grow as a researcher-engineer capable of advancing the development of domain-specific AI solutions that balance scalability, performance, and adaptability.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning Engineering 2022 – 2026 Sir M Visvesvaraya Institute of Technology (VTU Affiliated) GPA: 8.0/10.0

Coursework: C++, ML, AI, Python, Data Structures, Operating Systems, MySQL, DevOps, NLP, Git, and GitHub

INDEPENDENT RESEARCH

HyperForge: Adaptive Cross-Platform Game Engine for High-Performance Rendering *Ongoing* •

Developed HyperForge, a cross-platform game engine supporting Windows, Linux, and mobile platforms with high-performance rendering.

- Designed an adaptive rendering pipeline that dynamically adjusts graphics quality based on hardware capabilities, improving frame rates and visual fidelity.
- Implemented modular engine architecture with support for physics, asset management, and real-time rendering optimizations.

PROJECTS

Eye Tracking-Based Cursor Movement for Human-Computer Interaction *Ongoing* Team Size:- 4 ; Role: Team Lead

- Aim: To develop a system that enables users to control the computer cursor using eye movements, improving accessibility and human-computer interaction.
- Contribution: Implemented real-time eye-tracking using computer vision algorithms; designed gaze-to-cursor mapping logic; optimized performance for smooth and accurate cursor control.
- Outcome: Achieved cursor control with ~85% accuracy in stable lighting conditions, demonstrating potential for assistive technologies.
- Tools & Software: Python, OpenCV, Dlib, PyAutoGUI..

Traffic-violation-detection Using YoloV3 *November 2024* Team Size:- 4 ; Role: Team Lead

- Aim: To build an automated surveillance system for detecting traffic violations such as signal jumping and helmet-less riding using object detection.
- Contribution: Trained and fine-tuned YOLOv3 on a custom dataset of traffic scenarios; integrated detection pipeline with video streams; implemented violation logging and alert system.
- Outcome: Reached ~90% detection accuracy on test footage; system successfully detected and logged traffic violations in real-time video input.
- Tools & Software: Python, YOLOv3, TensorFlow/Keras, OpenCV.

Memory Leak Logger - C++ *October 2024*

- Aim: To design a lightweight tool to detect and log memory leaks in C++ applications for debugging and performance optimization.
- Contribution: Wrote custom memory allocation wrappers; implemented logging for dynamic allocations and deallocations; tested the tool with sample C++ programs to validate accuracy.
- Outcome: Successfully identified and logged memory leaks with minimal overhead, reducing debugging time by ~30% in test scenarios.
- Tools & Software: C++, Standard Template Library (STL), Valgrind (for validation).

Pothole Detection and Smart Navigation *March 2024 Team Size:- 2; Role: Team Member*

- Aim: To create a system capable of detecting road potholes and assisting vehicles with smart navigation for safer driving.
- Contribution: Processed road images and video streams using computer vision techniques; implemented pothole detection using contour and texture analysis; integrated GPS mapping for rerouting
 - Outcome: Achieved ~88% detection accuracy under daylight conditions; the prototype navigation system suggested alternative routes successfully in simulated tests.
- Tools & Software: Python, OpenCV, TensorFlow/Keras, GPS API (Google Maps).

PUBLICATION

“**EyeCommand: Intelligent Gaze & Blink Interface**,” K Ujjwal Reddy, Karthik M, P. V. Koushik Reddy, Vishal Murali Pathapati, Soumya Patil, IEEE Conference

- This project implements an assistive computing system that enables users to control their computer cursor and perform click operations using only their eyes and a standard webcam.
- The system provides an affordable alternative to expensive infrared eye-tracking hardware (typically \$1000+) while maintaining practical accuracy and reliability.
- The complete open-source implementation enables reproducibility, community extension, and real-world deployment for individuals with motor impairments who require alternative computer interaction methods.

“**Genetic Algorithm-Based Path Planning Using Adaptive Operators**,” K Ujjwal Reddy, Ashwini M Gorte, *International Journal of Innovative Research in Technology*, 2025

- Designed and implemented a Genetic Algorithm-based path planning system for robots, enabling efficient navigation in dynamic grid/maze environments.
- Applied adaptive crossover and mutation operators to improve convergence speed and solution accuracy compared to traditional GA..
- Evaluated through simulation experiments, demonstrating improvements in search efficiency and adaptability.

SKILLSET

Languages: C++, C, Python, HTML, CSS, PostgreSQL, MySQL.

Frameworks: LangChain, TensorFlow, PyTorch, CUDA, RAG, Scikit-Learn, Jupyter.

Software & Services: Computer vision, Anaconda, HuggingFace, Visual Studio, GitHub.

Languages: English, Kannada, Hindi, Telugu.

CERTIFICATIONS

- **LLM Engineering: Master AI, Large Language Models & Agents (25.5 hrs), Udemy**, July 2025
- **The Complete Agentic AI Engineering Course (2025)(17 hrs), Udemy**, September 2025
- **Python Object Oriented Programming (OOP): Beginner to Pro (27 hrs), Udemy**, September 2025
- **C++ Data Structures & Algorithms + LEETCODE Exercises (10.5 hrs), Udemy**, September 2025
- **The Complete C Programming Bootcamp (25 hrs), Udemy**, April 2025

LEADERSHIP ROLES

- Served as class representative, demonstrating leadership and communication skills, 2022 - Present. ●
- Volunteered at TechHub SMVIT, Bengaluru, organising and managing a technical club of over 50 members and leading the college's annual technical fest, two internal hackathons, and workshops by industry experts, September 2024.
- Coordinated events for Susamskriti, the college cultural club, serving on the Event Management Team and as Cultural Club Coordinator, demonstrating leadership and organisational skills, April 2023 to December 2024.